

Roll No.....

MID SEMESTER EXAMINATION SEP - 2025

Name of the Program: : B.Tech Aerospace Engineering

Semester: 3rd

Name of the Course: Applied Mechanics

Course Code:TAS305

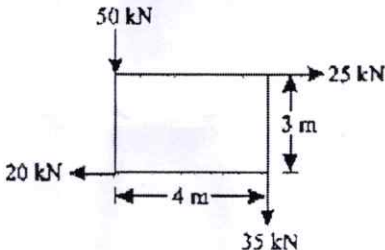
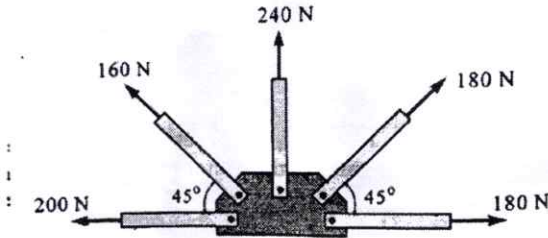
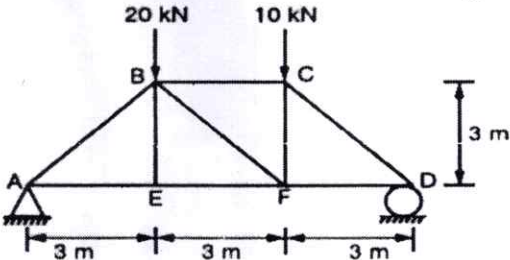
Time: 90 minutes

Maximum Marks: 50

Note:

Answer **all the questions** by choosing **any one of the sub questions**.

(i) Each question carries 10 marks

Q1	(10 marks)	
(a)	i. Explain classifications of force system. (3 Marks) ii. State and prove law of parallelogram of forces. (3 Marks) iii. State and prove Varignon's theorem. (4 Marks)	CO1
OR		
(b)	The resultant of the two forces, when they act at an angle of 60 degree is 14 N. If the same forces are acting at right angles, their resultant is $\sqrt{137}$ N. Determine the magnitude of the two forces.	
Q2	(10 marks)	
(a)	A system of forces is acting at the corners of a rectangular block as shown in Figure below. Determine the magnitude and direction of the resultant force. 	CO1
OR		
(b)	A gusset plate of roof truss is subjected to forces as shown in Figure below. Determine the magnitude of the resultant force and its orientation measured counter clockwise from the positive x-axis. 	
Q3	(10 marks)	
(a)	Determine the support reactions for the truss shown in Figure below. 	CO3
OR		

(b)	A simply supported beam AB of span 6 m is loaded as shown in Figure below. Determine the reactions at A and B.	
Q4	(10 marks)	
(a)	Determine the Centroid of the shaded area as shown in Figure below.	CO2
OR		
(b)	Determine the centroid of the area shown in Figure below with respect to the axis shown.	
Q5	(10 marks)	
(a)	Determine the Moment of Inertia about its Centroidal axes as shown in Figure below.	CO2
OR		
(b)	Find the moment of inertia about the centroidal axis of a channel section 100 mm × 50 mm × 15 mm as shown in figure below.	